QUESTION PAPER CODE : IRAU

Name of the Candidate: **Reg. No.**

APRIL 2025 REGULAR & ARREAR EXAMINATIONS
B.E/B.TECH. DEGREE
(Common to All Branches)
23PH120 PHYSICS

Duration : 3 Hours **Maximum: 100 Marks**

Answer ALL Questions

PART – A (10 X 2 =20)

		CO	Marks
A1.	Recall vectors and unit vectors in Cartesian and Polar coordinate systems.	CO1	***(2)***
A2.	Define Constraints and list the types of constraints.	CO1	***(2)***
A3.	Give the expression for de- Broglie wavelength in particle waves.	CO2	***(2)***
A4.	Cite the differences between classical and quantum computing.	CO2	***(2)***
A5.	State the three laws of thermodynamics.	CO3	***(2)***
A6.	List the properties of electromagnetic waves.	CO4	***(2)***
A7.	Define Poynting vector and mention its significance.	CO4	***(2)***
A8.	Recall population inversion and list the methods to achieve population inversion.	CO5	***(2)***
A9.	Recite the principle of Quantum Cascade Laser (QCL).	CO5	***(2)***
A10.	Define attenuation and mention the types of losses in optical fibers.	CO6	***(2)***

PART – B (3 x10 =30)

		CO	Marks
B1.	Illustrate the working of petrol engine with suitable P-V diagram and hence derive an expression for its efficiency.	*CO3*	***(10)***
B2.	Demonstrate the construction and working of Nd:YAG Laser and explain its merits, demerits and applications.	*CO5*	***(10)***
B3.	Explain the classification of optical fibers based on material, mode and refractive index.	*CO6*	***(10)***

PART – C (5 x10 =50)

			CO	Marks
C1.	***a.***	Demonstrate the transformation of scalars and vectors under rotation operation.	*CO1*	***(8)***
	b.	Establish the relation between vector components when the frame of reference is rotated through an angle of 30 degrees with respect to z –axis.	*CO1*	***(2)***
		(OR)		
C2.	***a.***	Deduce the expressions for velocity, acceleration and Newton's second law in cylindrical polar coordinate system.	*CO1*	***(8)***
	b.	Convert $x^2 + y^2 = 100$ into polar coordinates.	*CO1*	***(2)***
C3.	***a.***	Derive expressions for Schroedinger time dependent and time- independent wave equations for a particle wave.	*CO2*	***(8)***
	b.	Calculate the wavelength of an electron moving with a velocity of 10^6m/s.	*CO2*	***(2)***
		(OR)		
C4.	***a.***	Solve Schroedinger's wave equation for particle in a one dimensional box and obtain the energy eigen values and eigen function.	*CO2*	***(8)***
	b.	Calculate the 1^{st} excited state energy level of an electron confined in a nanowire of length 10 nm.	*CO2*	***(2)***

Contd…2…

C5.	***a.***	Let dQ be the amount of heat supplied to a perfect gas of mass 1 gm occupying a volume V at tempertaure T and pressure P. Derive the entropy of the perfect gas in terms of (i) temperature and pressure (ii) temperature and volume and (iii) pressure and volume using first law of thermodyanamics.	*CO3*	***(8)***
	b.	Calculate the increase in entropy when 5 kg of water at 100 $^\circ$C changes to steam.	*CO3*	***(2)***
		(OR)		
C6.	***a.***	Demonstrate the change in entropy in reversible and irreversible processes	*CO3*	***(8)***
	b.	Find the efficiency of Carnot's engine working between the steam point and ice point.	*CO3*	***(2)***
C7.	***a.***	Deduce Maxwell's equations in differential and integral forms.	*CO4*	***(8)***
	b.	If the amplitude of magnetic field is 3×10^{8} T, find the amplitude of electric field.	*CO4*	***(2)***
		(OR)		
C8.	***a.***	Derive the electromagnetic wave equation in free space and hence deduce the velocity of light in terms of permittivity and permeability of free space.	*CO4*	***(8)***
	b.	If the relative permittivity and relative permeability of a medium are 2.2 and 1.0, respectively. Find the velocity of the electromagnetic wave in the medium.	*CO4*	***(2)***
C9.	***a.***	Illustrate the principle, construction and working of optical fiber based Endoscope.	*CO6*	***(8)***
	b.	An optical signal with input power of 80 mW is injected into a fiber and the outcoming signal has a power of 40 mW. Deduce the power loss in decibels.	*CO6*	***(2)***
		(OR)		
C10.	***a.***	Derive an expression for the acceptance angle of an optical fiber in terms of refractive indices of core, cladding and fractional change in index.	*CO6*	***(8)***
	b.	Compute the numerical aperture of an optical fiber made of core and cladding with refractive indices 1.56 and 1.52, respectively.	*CO6*	***(2)***

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